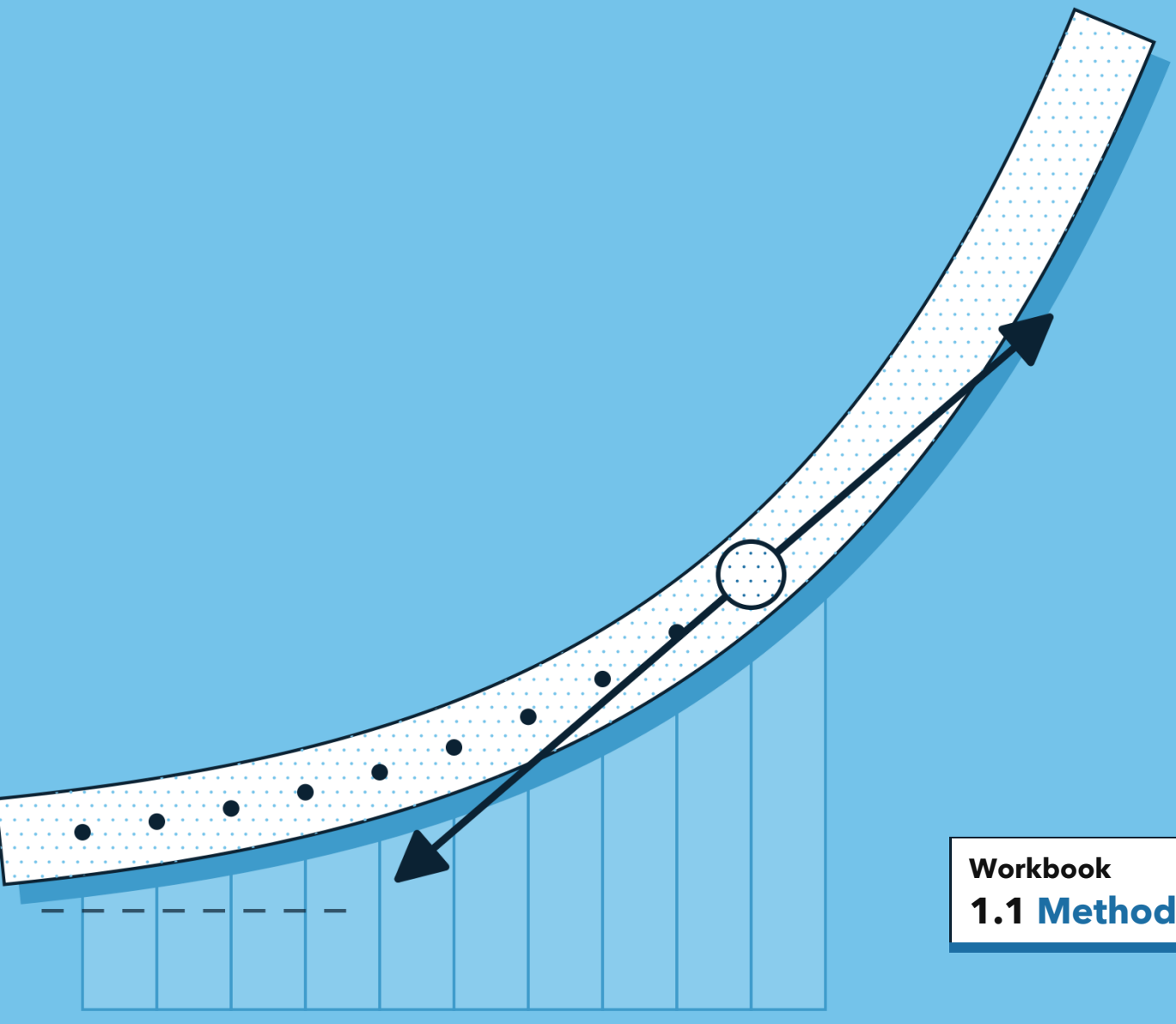
Vantage Tutoring

Mathematical Methods 3&4

1.1 Discovering e

Unit 3 · Calculus of Exponential and Logarithmic Functions · Year 12, 2027



Workbook
1.1 Methods

The Methods 3&4 series

30 sets, one road: from your first lesson to the external exam.



UNIT 3 · TOPIC 1

Calculus of exponential and logarithmic functions

- 1.1 **Discovering e** — You are here
- 1.2 Derivatives of e^x and ln x
- 1.3 Differentiating logarithms

TOPIC 2

Differentiation of trigonometric functions

- 2.1 Derivatives of sin and cos
- 2.2 Chain, product and quotient rules

TOPIC 3

Further applications of differentiation

- 3.1 The second derivative
- 3.2 Curve sketching and optimisation

TOPIC 4

Introduction to integration

- 4.1 Anti-differentiation
- 4.2 Initial conditions and motion

TOPIC 5

Discrete random variables

- 5.1 Discrete random variables
- 5.2 Bernoulli and binomial
- 5.3 Binomial mean and variance

UNIT 4 · TOPIC 6

Further integration

- 6.1 Area and approximating sums
- 6.2 The fundamental theorem
- 6.3 Area between curves

TOPIC 7

Trigonometry

- 7.1 Sine and cosine rules
- 7.2 Area and applications

TOPIC 8

Continuous random variables and the normal distribution

- 8.1 Continuous random variables
- 8.2 Mean and variance
- 8.3 The normal distribution

TOPIC 9

Sampling and proportions

- 9.1 Random sampling
- 9.2 Sample proportions
- 9.3 Normality of sample proportions

TOPIC 10

Interval estimates for proportions

- 10.1 Confidence intervals
- 10.2 Margin of error and sample size

FINISH

Exam preparation

- R.1 Unit 4 review for IA3
- R.2 Calculus revision
- R.3 Statistics revision
- R.4 Mock exam
- R.5 Targeted revision

NEXT STEP

Term 4 Exam Intensive Masterclasses

Available once you complete all 30 sets.

vantagetutoring.com.au

WHAT'S IN THIS SET

Set 1.1 Discovering e

Mathematical Methods 3&4 · Unit 3 · Calculus of Exponential and Logarithmic Functions · 2-hour session

Workbook Work through together in class — Pages 3 – 25

W	**Retrieval warm-up** Start here: a quick recap from earlier units	p. 3
A	**Discovering e** Compound interest · Euler's number e	p. 3-6
B	**The gradient of $y = a^x$** First principles · Why e is special	p. 7-10
C	**The graph of $y = e^x$** Key features · Transformations	p. 10-12
D	**The derivative of e^x** Tangents and normals · Exponential equations	p. 13-15
E	**Exam Practice** Mixed exam-style questions	p. 16-18
+	**Extension questions** Stretch problems	p. 19-20
★	**Challenge questions** Two extended CU problems	p. 21-25

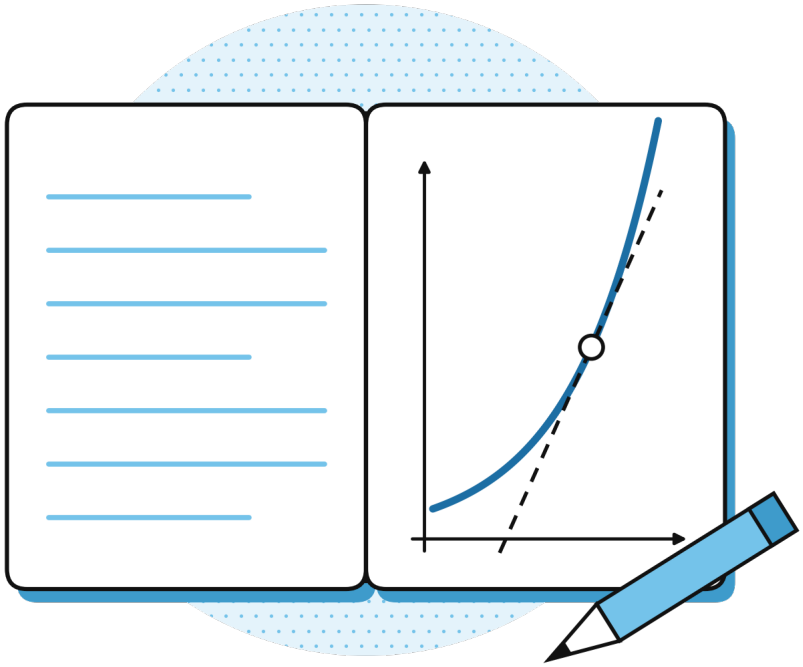
Homework Complete before your next session — Pages 27 – 33

TF	**Tech-free questions** No calculator	p. 27-30
TA	**Tech-active questions** Calculator allowed	p. 30-33

Learning objectives Tick each one when you can do it on your own

- ☐ Estimate the limit of $\frac{a^h-1}{h}$ as $h \to 0$ for different values of $a > 0$, using technology.
- ☐ Recognise that e is the unique number a for which this limit is 1, and that $e \approx 2.718$.
- ☐ Recognise the graph of $y = e^x$, its key features and simple transformations of it.
- ☐ Establish and use $\frac{d}{dx}(e^x) = e^x$, including for tangents and normals.

Objectives from the QCAA syllabus: Mathematical Methods 2025 v1.3, Unit 3 Topic 1.



Workbook

Work through these together in class

Retrieval Warm-up (from Year 11)

Question 1 (3 marks) **(TF) (SF)**

a. Differentiate $y = 2x^3 - 5x + 1$. (1 mark)

b. Find the gradient of $y = x^2 + 3x$ at the point where $x = 2$. (1 mark)

c. Simplify $\frac{2^{x+3}}{2^x}$. (1 mark)

Section A: Discovering *e* (4 Marks)

Compound interest and Euler's number

Exponential models describe bank balances, populations, drug doses and cooling. One base, the number e, makes the calculus of all of them simpler. This week we find e and see why it is special.

What happens to an investment when interest is added more and more often?

Compound Interest

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

- P is the principal (the amount invested) and A is the amount after t years.
- r is the annual interest rate written as a decimal, so 8% becomes 0.08.
- n is the number of times interest is added (compounded) each year.

Space for Personal Notes

Question 2 Walkthrough

Suppose \$1 is invested for one year at the (unrealistic) rate of 100% p.a., so $P = 1$, $r = 1$ and $t = 1$.

a. Find the value after one year if interest is compounded annually ($n = 1$).

b. Find the value after one year if interest is compounded every six months ($n = 2$).

c. Use your calculator to complete the table, giving each value to 6 decimal places.

Compounded	n	Expression	Value after 1 year (\$)
Annually	1	$\left(1+\frac{1}{1}\right)^1$	2.000000
Quarterly	4		
Monthly	12		
Daily	365		
Hourly	8760		
Every minute	525 600		

d. Describe what happens to the value as n gets larger.

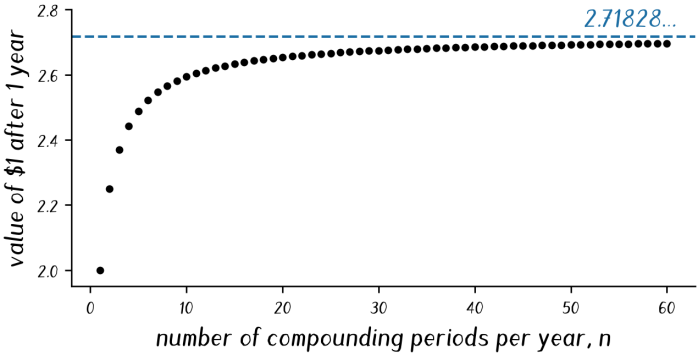
CALCULATOR Evaluate $\left(1+\frac{1}{n}\right)^n$ for each n

Casio fx-CG50 AU: `Run-Matrix`: type `(1+1÷4)^4`, `EXE`; press ◀ to edit n, then `EXE` again

TI-84 Plus CE: type `(1+1/4)^4`, `ENTER`; `2nd > ENTER` (ENTRY) brings it back to edit n

Euler's Number, e

$$e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = 2.71828\ldots$$



- Like π, e is an _____ number: its decimals never end or repeat.
- When interest is added continuously (every instant), the amount is $A = Pe^{rt}$.

Space for Personal Notes

Have a go at the following question!

Question 3 (4 marks) **(TA) (SF)**

Mei invests $6500 for 5 years at 8.4% p.a.

CALCULATOR

Casio fx-CG50 AU: `MENU > Run-Matrix`, type the expression, `EXE`

TI-84 Plus CE: home screen, type the expression, `ENTER`

a. Find the value of the investment if interest is compounded monthly. (2 marks)

b. Find the value of the investment if interest is compounded continuously. (1 mark)

c. How much more does Mei earn with continuous compounding? (1 mark)

NOTE: Round money to the nearest cent only at the end. Rounding $\frac{r}{n}$ early changes the final answer.

Section B: The gradient of $y = a^x$ (1 Mark)

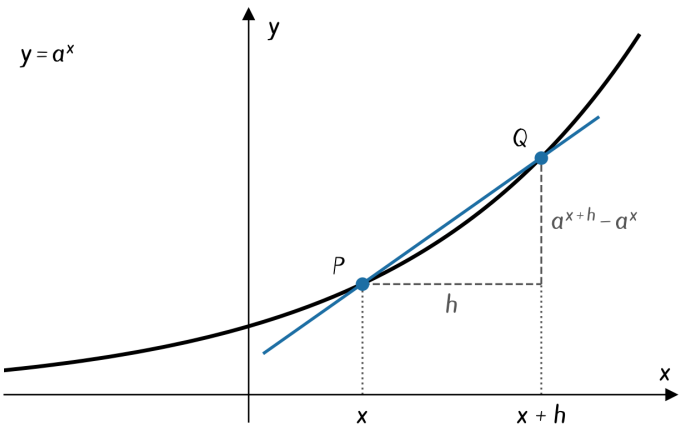
First principles

Is there an exponential function whose gradient function is itself?

Gradient of an Exponential Function

The gradient of the chord PQ approaches the gradient of the tangent at P as $h \to 0$:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$



- For $f(x) = a^x$: $f'(x) = \lim_{h \to 0} \frac{a^{x+h} - a^x}{h} = \lim_{h \to 0} \frac{a^x(a^h - 1)}{h} = a^x \times \lim_{h \to 0} \frac{a^h - 1}{h}$
- So the gradient of $y = a^x$ is always a constant multiple of _____.

Question 4 Walkthrough

Use $h = 0.001$ to estimate $\frac{a^h - 1}{h}$ for each value of a. Give answers to 4 decimal places.

a	2	2.5	2.7	2.72	3
$\frac{a^h - 1}{h}$	0.6934				

a. Complete the table.

b. Between which two values of a in the table does $\frac{a^h - 1}{h}$ equal 1?

c. Now use $h = 0.0001$ and trial and improvement to find, correct to 2 decimal places, the value of a for which $\frac{a^h - 1}{h} = 1$.

CALCULATOR Evaluate $\frac{a^h - 1}{h}$

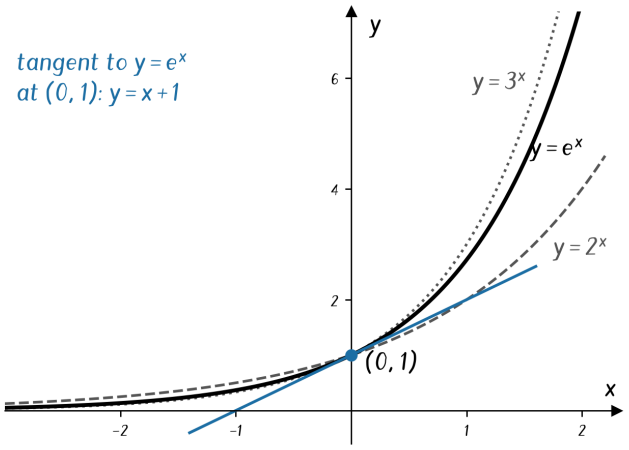
Casio fx-CG50 AU: `Run-Matrix`: type `(2.5^0.001−1)÷0.001`, `EXE`; press ◀ to change a (or h), then `EXE`

TI-84 Plus CE: type `(2.5^0.001−1)/0.001`, `ENTER`; `2nd > ENTER` (ENTRY) to change a (or h)

Why e Is Special

e is the only base for which $\lim_{h\to 0}\frac{e^h-1}{h}=1$. Putting this into the result above:

$$\frac{d}{dx}(e^x)=e^x$$



- $y=e^x$ is its own gradient function: at every point, gradient = y-coordinate.
- At (0, 1) the tangent to $y=e^x$ has gradient exactly 1. For 2^x it is less than 1, and for 3^x it is more than 1.

Question 5 (1 mark) **(TA) (SF)**

The graph of $y=a^x$ crosses the y-axis with a gradient of 1.5. The value of a is closest to

CALCULATOR

Casio fx-CG50 AU: `MENU > Run-Matrix`, type the expression, `EXE`

TI-84 Plus CE: home screen, type the expression, `ENTER`

A. 1.5

B. 3.0

C. 4.1

D. 4.5

TIP: This limit is not on the formula sheet. In Set 1.3 you will see that it equals $\ln a$.

Space for Personal Notes

Section C: The graph of $y = e^x$ (10 Marks)

Key features and transformations

What does the graph of $y = e^x$ look like, and how do transformations move it?

Key Features of $y = e^x$

- Domain: _____ Range: _____
- y-intercept: _____ Horizontal asymptote: _____
- The graph is always increasing and always concave up.
- For $y = ae^{x-h} + k$: a dilates (and reflects if $a < 0$), h shifts right, k shifts up and moves the asymptote to $y = k$.

Question 6 (6 marks) **(TF) (SF)**

For each function, describe the transformation(s) of $y = e^x$, then state the equation of the asymptote and the coordinates of the y-intercept.

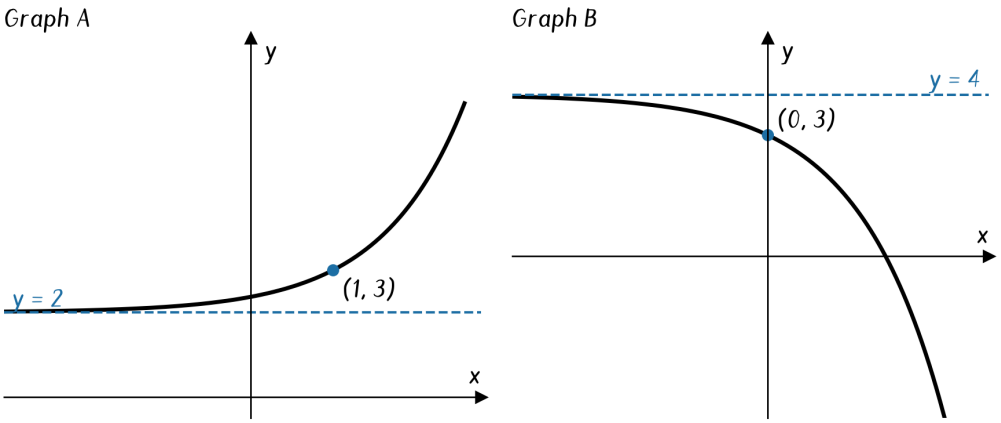
a. $y = e^x - 3$ (2 marks)

b. $y = 1 - e^x$ (2 marks)

c. $y = e^{x+2}$ (2 marks)

Question 7 (4 marks) **(TF) (SF)**

Each graph below is a transformation of $y = e^x$. Asymptotes are dashed.



a. Graph A has an equation of the form $y = e^{x-h} + k$. Find h and k. (2 marks)

b. Graph B has an equation of the form $y = ae^x + k$. Find a and k. (2 marks)

NOTE: Read the asymptote first. It gives k straight away, then one point finds the last unknown.

Space for Personal Notes

Section D: The derivative of e^x (12 Marks)

Tangents, normals and equations

How do we differentiate expressions that contain e^x?

Derivative Rules

- $\frac{d}{dx}(e^x) = e^x$ and $\frac{d}{dx}(ke^x) = ke^x$ for any constant k.
- Differentiate sums and differences term by term, e.g. $\frac{d}{dx}(x^2 + 3e^x) = 2x + 3e^x$.
- Index laws help: $e^{x+c} = e^c e^x$, so $\frac{d}{dx}(e^{x+c}) = e^{x+c}$.
- Tangent at $(a, f(a))$: $y - f(a) = f'(a)(x - a)$. The normal has gradient $-\frac{1}{f'(a)}$.

Question 8 Walkthrough

Consider the curve $y = e^x$.

a. Find the gradient of the curve at $x = 1$.

b. Find the equation of the tangent at $x = 1$.

c. Show that this tangent passes through the origin.

Discuss: Why do scientists usually choose base e rather than base 2 or base 10 for growth models?

Have a go at the following questions!

Question 9 (4 marks) **(TF) (SF)**

Differentiate each of the following with respect to x.

a. $y = 4e^x$ (1 mark)

b. $y = x^3 - 2e^x + 7$ (1 mark)

c. $f(x) = \frac{e^x + 6x}{3}$ (1 mark)

d. $y = e^{x+3}$ (1 mark)

Question 10 (3 marks) **(TF) (SF)**

Determine the equations of the tangent and the normal to the curve $y = e^x + 1$ at the point where $x = 0$.

Question 11 (3 marks) **(TF) (SF)**

Solve each equation for x.

a. $e^{2x+1} = e^{x-4}$ (1 mark)

b. $e^{x^2} = e^{4x-3}$ (2 marks)

Question 12 (2 marks) **(TF) (CF)**

The point P lies on the curve $y = 2e^x$ and has y-coordinate 6. Without finding the x-coordinate of P, find the gradient of the curve at P. Explain your reasoning.

NOTE: The normal's gradient is $-\frac{1}{f'(a)}$, not $-f'(a)$. Keep exact values such as e until the final line.

Space for Personal Notes

Section E: Exam Practice (13 Marks)

Exam conditions: about 20 minutes, no notes. Use your calculator only on (TA) questions.

Question 13 (4 marks) **(TF) (SF)**

The graph of $f(x) = e^{x-1} + k$ passes through the point (1, 4).

a. Determine the value of k. (1 mark)

b. Determine the exact coordinates of the y-intercept of the graph. (1 mark)

c. Determine the equation of the tangent to the graph at the point (1, 4). (2 marks)

Question 14 (4 marks) **(TA) (CF)**

Priya invests \$8000 at 5.6% p.a. compounded continuously. Her brother Tom invests \$8000 at r% p.a. compounded quarterly. After 6 years their two investments are worth the same amount.

CALCULATOR

Casio fx-CG50 AU: `MENU > Graph`, `SHIFT > V-Window`, `SHIFT > G-Solv > INTSECT`, `MENU > Table`

TI-84 Plus CE: `2nd > CALC > intersect`, `2nd > TBLSET`, `2nd > TABLE`

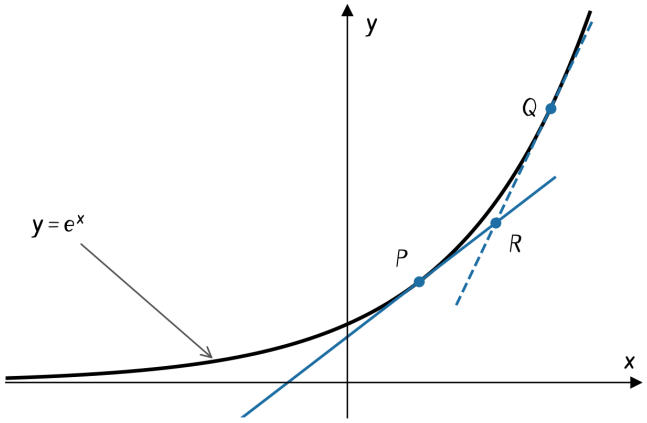
a. Determine the value of Priya's investment after 6 years. (1 mark)

b. Determine the value of r, correct to 2 decimal places. (2 marks)

c. Determine the number of whole years it takes for Priya's investment to first exceed \$12 000. (1 mark)

Question 15 (5 marks) **(TF) (CU)**

The points P and Q lie on the curve $y = e^x$, with Q exactly 1 unit to the right of P. The tangents to the curve at P and at Q meet at the point R, as shown. No coordinates are given.



Show that, wherever P is on the curve, R is always the same horizontal distance to the right of P, and the height of R is always the same multiple of the height of P. State both constants exactly. Hence determine the exact coordinates of R when P is the point (0, 1).

Extension Questions

Question 16 (5 marks) **(TF) (CU)**

The line $y = 3x$ and the curve $y = e^x + k$, where k is a constant, are drawn on the same axes. Changing k slides the curve up or down. For exactly one value of k the line touches the curve at a single point, so that the line is a tangent to the curve there.

Determine this value of k and the point of contact, giving exact values. Hence deduce, with justification, the values of k for which the line and the curve do not meet.

Question 17 (4 marks) **(TF) (CF)**

The curves $y = e^{2x}$ and $y = 4e^x - 3$ meet at two points, A and B. (Hint: $e^{2x} = (e^x)^2$.)

a. Find the exact coordinates of A and B. (3 marks)

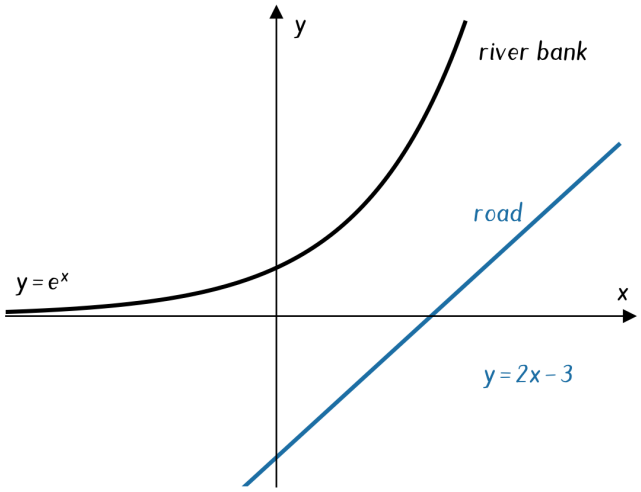
b. Find the gradient of the curve $y = 4e^x - 3$ at A and at B. (1 mark)

Challenge Questions

Two harder problems for students who want a real test. Show all reasoning; a full page is given for each.

Question 18 (7 marks) **(TF) (CU)**

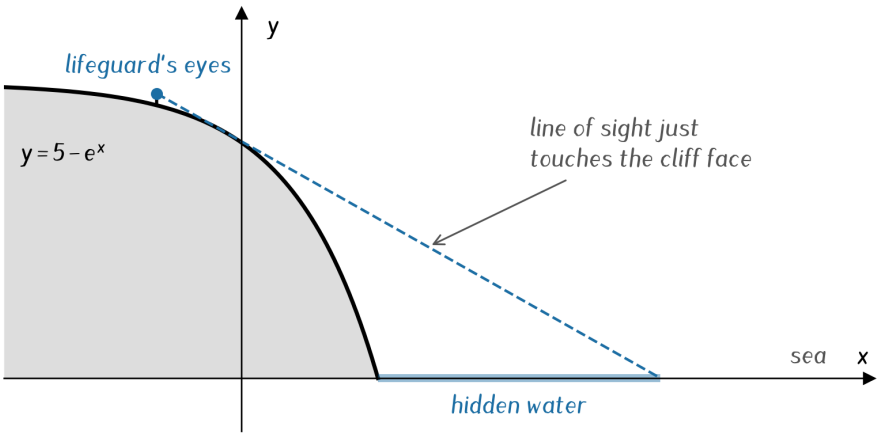
A straight road and a river bank are drawn on a map with axes in kilometres. The road follows the line $y = 2x - 3$ and the river bank follows the curve $y = e^x$, as shown. The council will build the shortest possible straight footpath joining the river bank to the road, and has budgeted for a path of at most 1.6 km.



Determine the exact coordinates of both ends of the shortest footpath, justifying why no shorter path exists, and find its exact length. Hence decide whether the path is within budget. (Use $log_e 2 \approx 0.69$.)

Question 19 (8 marks) **(TA) (CU)**

The cross-section of a rounded sea cliff is modelled by $y = 5 - e^x$, $x \leq log_e 5$, where x and y are in tens of metres and the sea is the x-axis, so the base of the cliff is at $(log_e 5, 0)$. A lifeguard stands on a lookout on the clifftop where $x = -1$, with her eyes 2 m above the ground. The cliff face bulges outwards, so the water close to its base is hidden from her: her line of sight to the nearest water she can see just touches the cliff face, as shown. Swimmers are allowed in a zone from 20 m to 60 m out from the base of the cliff.



Determine how far from the base of the cliff the nearest water she can see is, to the nearest metre, and decide whether she can see the whole swimming zone. The council could move the lookout forward to $x = -0.5$, or keep it at $x = -1$ and build a platform that raises her eyes to 3 m above the ground. Recommend one option, justifying your choice with calculations.

CALCULATOR

Casio fx-CG50 AU: `MENU > Graph`, `SHIFT > V-Window`, `SHIFT > G-Solv > INTSECT`

TI-84 Plus CE: `2nd > CALC > intersect`

Extra Working Space

Extra Working Space

Self-check

Tick each statement you can now do on your own. Anything unticked is the first thing to ask about next session.

- ☐ I can explain where e comes from using compound interest.
- ☐ I can estimate $\lim_{h \to 0} \frac{a^h - 1}{h}$ and explain why e is the base that makes it 1.
- ☐ I can describe and sketch transformations of $y = e^x$, with asymptote and intercept.
- ☐ I can differentiate expressions containing e^x and find tangents and normals.
- ☐ I can solve equations of the form $e^{f(x)} = e^{g(x)}$.



Homework

Complete before your next session

Section A: Tech-Free Questions (15 Marks)

Simple Familiar

Question 20 (1 mark) **(TF) (SF)**

Which statement about the graph of $y = 3 - e^x$ is correct?

A. Its horizontal asymptote is $y = -3$.

B. Its y-intercept is (0, 3).

C. Its range is $y < 3$.

D. It is increasing for all x.

Question 21 (4 marks) **(TF) (SF)**

Differentiate each of the following with respect to x.

a. $y = 5e^x - x^2$ (1 mark)

b. $y = \frac{2}{3}e^x + \frac{1}{x}$ (1 mark)

c. $y = e^{x-2} + e^2$ (1 mark)

d. $y = \frac{e^{2x}+e^x}{e^x}$ (1 mark)

Question 22 (3 marks) **(TF) (SF)**

Solve each equation for x.

a. $e^{3x} = e^{x+8}$ (1 mark)

b. $e^{x} \times e^{x+1} = e^{7}$ (2 marks)

Question 23 (3 marks) **(TF) (SF)**

Find the equation of the tangent to $y = 3e^{x} - x$ at the point where $x = 0$.

Complex Familiar

Question 24 (4 marks) **(TF) (CF)**

a. Show that the tangent to $y = e^x$ at $x = a$ crosses the x-axis at $x = a - 1$. (3 marks)

b. Hence state where the tangent at $x = -2$ crosses the x-axis. (1 mark)

Section B: Tech-Active Questions (15 Marks)

Simple Familiar

Question 25 (3 marks) **(TA) (SF)**

Lena has $10 000 to invest for 4 years. Bank A pays 6% p.a. compounded monthly. Bank B pays 5.95% p.a. compounded continuously.

> **CALCULATOR**
> **Casio fx-CG50 AU:** `MENU > Run-Matrix`, type the expression, `EXE`
> **TI-84 Plus CE:** home screen, type the expression, `ENTER`

a. Find the value of Lena's investment with Bank A. (1 mark)

b. Find the value of Lena's investment with Bank B. (1 mark)

c. Which bank should Lena choose, and how much more will she have with it? (1 mark)

Question 26 (3 marks) **(TA) (SF)**

Find the coordinates of the point on the curve $y = 2e^x - 3x$ where the tangent is horizontal. Give values correct to 2 decimal places.

CALCULATOR

Casio fx-CG50 AU: `MENU > Graph`, `SHIFT > V-Window`, `SHIFT > G-Solv > MIN`

TI-84 Plus CE: `2nd > CALC > minimum`

Complex Familiar

Question 27 (4 marks) **(TA) (CF)**

CALCULATOR

Casio fx-CG50 AU: `MENU > Run-Matrix`, type the expression, `EXE`

TI-84 Plus CE: home screen, type the expression, `ENTER`

a. Use $h = 0.0001$ to estimate $\lim_{h \to 0} \frac{4^h - 1}{h}$ correct to 3 decimal places. (1 mark)

b. Hence estimate the gradient of $y = 4^x$ at $x = 1.5$. (1 mark)

c. By trial and improvement, find the value of a, correct to 2 decimal places, for which $\lim_{h \to 0} \frac{a^h - 1}{h} = 2$. (2 marks)

Complex Unfamiliar

Question 28 (5 marks) **(TA) (CU)**

A chord joins the point (0, 1) to the point (b, e^b) on the curve $y = e^x$, where $b > 0$. Between these two points there is a point on the curve, with x-coordinate c, where the tangent is parallel to the chord. Ana claims that this point is always more than halfway along, that is, $c > \frac{b}{2}$.

Investigate Ana's claim for $b = 1$, $b = 2$ and $b = 4$, giving each value of c correct to 2 decimal places. Evaluate the claim, and explain, using the shape of the graph of $y = e^x$, why it holds.

CALCULATOR

Casio fx-CG50 AU: `MENU > Run-Matrix`, type the expression, `EXE`

TI-84 Plus CE: home screen, type the expression, `ENTER`

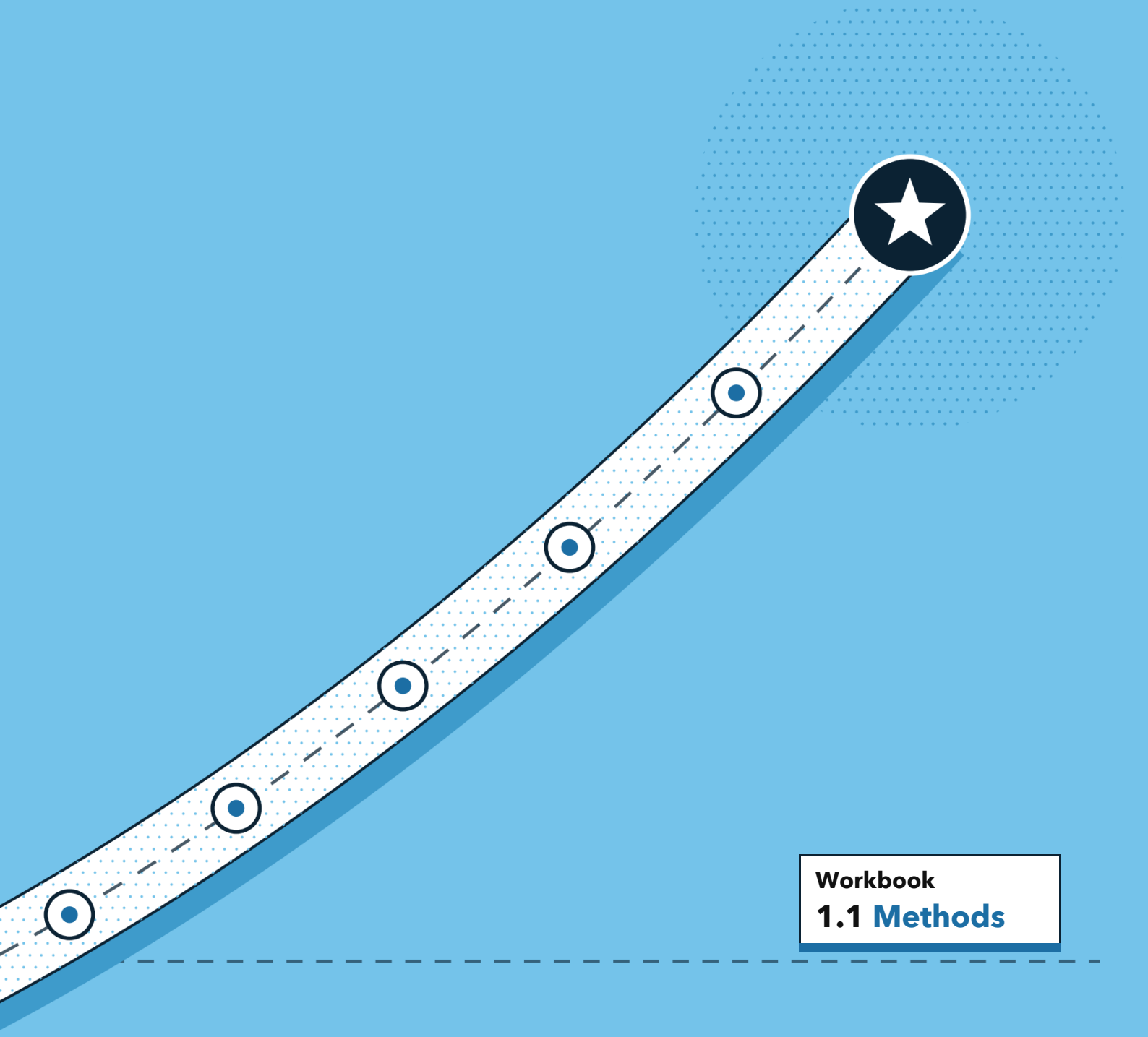
Extra Working Space

Extra Working Space

Vantage Tutoring

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Mathematical Methods 3&4 · Year 12, 2027



Workbook

1.1 Methods